
Annex C (Informative): Example mechanism for setting base and single shift 7-bit language tables

C.1 Introduction

This annex gives an overview on how the alphabet locking shift and single shift mechanism of the 7-bit alphabets work. This annex shows how a message with an alphabet indication of the European languages (Latin script) (0x10) Alphabet Identifier is decoded, but the same principles apply to other alphabets. Note that the locking shift and single shift alphabets must be set to the same value (that there are two settings is a remnant from earlier versions).

C.2 Example of setting 7-bit alphabet language tables

This example outlines the behaviour of both supporting and non-supporting receiving entities where the Thai National Language Single Shift Table (0x1D) **or** the Thai National Language Locking Shift Table (or both) is indicated in the received message. Note that setting either the Single Shift table or the Locking shift table, since the reference code (0x1D) is in the 0x10 or larger range, will set all three (Base, SS2, SS3) tables. If they are explicitly set inconsistently, it is same as not setting them.

A non-supporting receiving entity will ignore the National Language IEs and decode the message contents using the 7-bit default alphabet tables (0x00) defined in subclause 6.2.1 (except 6.2.1.3), including possible SS2 characters to the 7-bit default alphabet extension table specified in subclause 6.2.1.2. This will for the most part result in completely garbled text, but such things as USSD commands (using *, +, #, 0-9) and 2-letter language tags (using lowercase a-z) will remain unchanged; likewise for much of English language text *in lowercase* and some simple punctuation. A non-supporting receiving entity may use the 0x10 alphabet as fallback instead of 0x00, if the 0x10 alphabet (European languages) is supported.

A receiving entity that supports the Thai Language Tables will detect a National Language IE in a TP User Data Header. This IE tells the receiving entity that the Thai Alphabet (all three tables, locking shift, SS2, SS3) is used. A supporting receiving entity will notice the language code, in this example coded as 0x1D, and therefore use the Thai National Language Tables defined in subclause A.4.1D instead of the 7 bit default alphabet extension table defined in subclause 6.2.1.

For 7-bit alphabets with reference number 0x10 or higher, submessages of a message need not use the same 7-bit alphabet, they may be mixed and may also mix in the UCS2 (UTF-16BE) alphabet for some of the submessages, but no 7-bit alphabet with reference number 0x00 (even as default) to 0x0F.

C.3 Void

